

Nuclear magnetic relaxation and diffusion study of the ionic liquids 1-ethyl- and 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide confined in porous glass

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Abstract

The molecular dynamics of the room-temperature ionic liquid 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (Bmim Tf₂N) confined in porous glass is studied by nuclear magnetic resonance (NMR) relaxometry and diffusometry and is compared with the bulk dynamics over a wide temperature range. The molecular reorientation processes for anions and cations alike are found to be significantly affected by the presence of the glass interface at high temperatures. In this respect, the ionic liquid behaves similarly to polar liquids where proton NMR relaxation is governed by reorientations mediated by translational displacements (RMTDs). This process becomes less significant towards lower temperatures when the characteristic translational correlation times of the ions approach a timescale comparable with those of the RMTD process, and the relaxation dispersions in bulk and in confinement become similar below a temperature corresponding to about $1.2T_g$, a value where the onset of dynamic heterogeneity has been observed before. The self-diffusion coefficient, on the other hand, is found to be strongly reduced than the bulk within the accessible temperature range of 248 K and above and is significantly slower than expected from the tortuosity effect, suggesting that ion-surface interactions affect the macroscopic properties.

KEYWORDS

diffusion, dynamics of confined liquids, ionic liquids, NMR relaxation, super cooled liquids

1 | INTRODUCTION

Following the growing interest in ionic liquids (ILs) in chemistry and engineering applications, research in the last decade has been expanded towards the peculiar behavior of these liquids as confined fluids and interacting with solid substrates. ILs in confinement are studied for a number of reasons, for example, as hybrid conductive materials combining the advantages of mechanical stability and controllable conduction properties or as part of solid-supported catalyst materials. In

all cases, an understanding of the deviations of microscopic and macroscopic properties of IL in comparison with the bulk is attempted. But also from a fundamental point of view, the behavior of ions in the presence of interfaces is of considerable interest. A summary of recent findings can be found in Hayes et al.^[1] and Zhang et al.^[2]

Water and a number of liquids with low viscosity, such as organic solvents with different polarity, have been studied in geometrical confinement for decades, and their behavior can be considered as reasonably well understood. They are subject to two concurring effects, the

presence of the surface and the geometrical aspect of restricted space itself. Whereas the latter affects thermodynamic properties such as melting point and glass transition temperatures, the former possesses the potential, depending on the nature of the solid surface, to enhance or reduce molecular order and to restrict or even promote mobility on a molecular scale.^[3] The transition from solid to flexible interfaces allows the improved understanding of biological systems such as water dynamics in cell membranes.

Nuclear magnetic resonance (NMR) has traditionally provided a most suitable noninvasive experimental access to confined fluids. At one end of the application spectrum, the fluid serves as a tracer that probes the pore space of the material of interest and allows the determination of surface-to-volume ratios, average pore and throat sizes, or connectivity, providing access complementary to the Brunauer–Emmett–Teller method or mercury adsorption but being more sensitive to surface chemistry. This has been used successfully for rocks, soils, catalyst carrier materials, and related problems in chemical engineering. On the other hand, a growing field of research addresses the details of interaction of the molecules with the interfaces as such; due to the inherently low sensitivity of NMR, this frequently requires the study of porous materials with a large internal surface area rather than flat, wafer-like sample geometries.

The two most widespread NMR applications for the study of confined fluids are diffusion and relaxation measurements, which are sensitive to different timescales and length scales. Pulsed field gradient (PFG) or constant field gradient techniques measure the mean-squared displacement of spin-bearing molecules, where averaging takes place over the spin ensemble and over timescales of typically 10^{-3} to 1 s or root-mean-squared displacements of 10^{-8} to 10^{-5} m in fluids depending on their viscosity. In microporous and mesoporous media, this often means that molecules sample many pores and therefore probe the representative elementary volume that covers all existing pore sizes in their statistical representation; diffusion then assumes its tortuosity limit, which can be verified by time-dependent diffusion measurements. The pore size of macroporous systems, falling in the range of the experimentally accessible root-mean-squared displacements, is accessible from this time dependence but is not the subject of this work.

On the other hand, relaxation measurements are sensitive to timescales on the order of the inverse Larmor frequency at which the experiment is carried out, often 10^{-9} s or less when conventional laboratory magnetic fields of several teslas are used; these timescales correspond to local motion such as rotation and short-distance translation on the order of the molecular size. However,

with the use of low-field and variable-field NMR relaxometry, it is possible to extend the range towards timescales of 10^{-4} s or even beyond, approaching the diffusion experiment range and thus allowing for a complete description of molecular mobility over many orders of magnitude.^[4] Relaxation measurements can be analyzed in such a way that both rotational and translational time constants are obtained.

Simple liquids are characterized by short correlation times in bulk, typically 10^{-10} to 10^{-11} s, frequently accessible only indirectly by applying the proper autocorrelation function for reorientation that links the observed relaxation times with molecular parameters. In many cases, confinement of these liquids in microporous and mesoporous materials was found to lead to dramatically prolonged correlation times, that is, the existence of very slow reorientational processes promoted by the existence of a surface. The interaction strength of the liquid with the surface has been identified as the critical parameter, with polar liquids on polar surfaces (such as untreated glass) showing very pronounced slowing down of their dynamics, and with hydrogen bonds probably playing an important part, but by no means being entirely responsible for the effect.^[5] This behavior has been discussed in terms of reorientations mediated by translational displacements (RMTDs),^[6] where it was shown for all studied cases of small molecules that fast exchange between the surface layer and the pore interior promotes the sampling of the surface by a process called bulk-mediated surface diffusion.^[6,7] The alternative description of bulk (three-dimensional) versus surface (two-dimensional) diffusion processes remains in discussion to this date. Peculiar to NMR relaxation is the fact that, apart from reorientation of the molecules themselves, further mechanisms such as paramagnetic relaxation with impurities on the surface, or close to the surface within the solid, can lead to similar or even dominating relaxation contributions that can be considered as an additive relaxation rate. In this case, paramagnetic centers act as relaxation sinks via dipolar interaction between the nucleus and the unpaired electron, where the relative motion of the spin-bearing molecule with respect to the fixed center is responsible for the frequency dependence of relaxation.^[8] Again, the way how molecules diffuse along the surface enters as an important parameter of this contribution, even though the paramagnetic impurities do not have an influence of the molecular dynamics per se.^[9]

Most of the typical room-temperature ILs often show significant relaxation dispersion even in bulk, which can be fully described by models taking rotational and translational motions into account. One advantage of NMR relaxometry is its possibility to separate these two contributions, and it has been employed in the understanding

of glass-forming liquids and corresponding ILs in the supercooled regime.^[10,11]

When the already relatively slow molecular dynamics of ILs is subjected to the presence of the further restrictions of a porous medium, one would, first of all, expect the occurrence of the same features, which were found in uncharged liquids, namely, a reduced macroscopic diffusivity due to motion within the tortuous solid matrix and the possible influence of the interface, which is more difficult to predict in detail. To this date, only a limited number of studies have investigated such systems by means of NMR methods, and only few works are available to verify the structural and dynamic processes found by numerical simulations for certain combinations of ILs and surfaces.

As is summarized in Hayes et al.^[11] and Ghoufi et al.,^[12] two concurring trends are predicted and even shown experimentally in the literature: While in many randomly structured materials the tortuosity effect dominates and self-diffusion coefficients of ILs are found to be smaller than in bulk, enhanced diffusivities have been reported as well. Ghoufi et al.^[12] stated that “Dynamics of confined RTILs remains unclear and is still a matter of debate.” In Iacob et al.,^[13] it is shown for the example of 1-hexyl-3-methylimidazolium hexafluorophosphate (Hmim PF₆) in untreated porous silicon membranes of 7.5-nm pore size that self-diffusion was reduced about 10-fold, but a substantially smaller reduction was observed after silanization of the surface, which was explained by the slowing-down effect of hydrogen bonds. Li et al.^[14] observed a reduction by a similar factor for 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (Bmim Tf₂N) in a 9.5-nm-pore-size silica material, compared with a less pronounced effect in simulations of porous carbon with somewhat smaller pores. This work highlights the importance of liquid–surface interaction, in particular the shape of the potential, and suggests different mobilities as a function of the wall distance, with the slowest molecules being found immediately at the wall. Pal et al.^[15] reported preferential orientations of Bmim cations of Bmim PF₆ IL on silica surfaces, which are relevant for the RMTD relaxation process.

In Filippov et al.,^[16] it was found that the diffusion decay of a large-molecule IL (phosphonium bis(salicylate)borate) in standard porous glass Vycor (4-nm pore diameter) show two components, one bulk-like, and the other up to 35 times more enhanced than the bulk but gradually disappearing with increasing encoding time. In this substance, the dynamics of anions and cations was shown to have the same temperature dependence. Relaxometry studies, on the other hand, were carried out not only on gel polymer electrolyte^[17] but also on other nonclassical porous media such as ionogels and

cellulose gels or Nafion membranes. In Petit et al.,^[18] power-law relations are found for anion and cation in Bmim Tf₂N and strongly correlated motions are concluded. In Neves et al.^[19] and Martins et al.,^[20] three different ILs are studied, among them Bmim BF₄, and the difference in ¹H relaxation compared with water is reported; they could successfully fit the dispersions by a combination of rotational and translational fast motions and RMTD with an equal weighting of wave numbers. A similar approach has been used for another IL, Bmim Cl, in a cellulose gel matrix.^[21] The study that is possibly most similar to this work is the investigation of Bmim octyl sulfate in silica pores by Kruk et al.^[22] where a comparison with bulk data is interpreted as a second, much slowed-down translational component of molecules close to the pore wall. For this interpretation, the expression for bulk intermolecular dipolar relaxation has been used.

Recently, the authors have presented a combined study of NMR relaxometry and diffusion measurements, where a deviation of the molecular dynamics from Arrhenius or Vogel–Fulcher–Tammann (VFT) behavior, respectively, has been found in the supercooling regime^[10] for the bulk IL 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (Bmim Tf₂N). In this work, the same IL is studied confined in Vycor porous glass with a nominal pore diameter of 4 nm, and the results are compared with the bulk behavior, providing an interpretation for the molecular dynamics under solid confinement.

2 | EXPERIMENTAL

Two dialkylimidazolium ILs with either ethyl or butyl alkyl side chains are investigated here: 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (Emim Tf₂N) and 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (Bmim Tf₂N). The discussion will focus on the latter due to the much wider temperature range available in the bulk liquid phase. The melting (T_m) and glass transition (T_g) temperatures of Bmim Tf₂N are 270.4 and 186 K, respectively.^[23] Both substances were purchased from Sigma Aldrich and were used without further purification. The IL samples were first placed in a vacuum oven at 403 K for at least 24 hr and subsequently purged with nitrogen flow. The ILs were then transferred into NMR tubes (10- and 5-mm outer diameter) and flame sealed immediately. High-resolution ¹H-NMR spectra (at 7 T) of these samples did not show any detectable peaks other than the main peaks assigned to the ILs. Cylindrical rods of Vycor porous glass (Corning Glass Works, OEM Sales Service, Box 5000, Corning, NY 14830) were prepared by boiling in 30%

aqueous H₂O₂ solution for 30 min followed by washing in bidistilled water and subsequently evacuating for 24 hr; this process was repeated at least twice. Dry glass samples were then immersed in the ILs and allowed to equilibrate for at least 24 hr. The samples were flame sealed into NMR tubes just as the bulk samples.

The longitudinal nuclear spin relaxation times T_1 of the ^1H nuclei of the cations and ^{19}F of the anions, respectively, were measured at Larmor frequencies between few kHz and 40 MHz employing a fast field-cycling^[24] NMR relaxometer (Stelar s.r.l., Mede, Italy) by integrating the free induction decay signal at the detection field. All relaxation curves could be fitted by monoexponential functions. The Carr–Purcell–Meiboom–Gill pulse sequence^[6] was used to measure T_2 relaxation time and the stimulated-echo PFG NMR pulse sequence^[6] was used to measure the self-diffusion coefficients, both on a 7-T Bruker Avance III spectrometer operating at 300-MHz ^1H resonance frequency, equipped with a Diff-30 diffusion probe with a maximum gradient of 12 T/m. The temperature for both NMR techniques was controlled either by a precooled stream of dry air or by evaporated gas from a liquid nitrogen reservoir within an accuracy of 0.2 K.

3 | THEORETICAL BACKGROUND

For two magnetic field gradient pulses of identical strength g , duration δ , and separation Δ (measured between centers of the pulses), the signal attenuation due to diffusion of the acquired echo is given by

$$A(\Delta, \delta, g) = A_0 e^{-(\gamma g \delta)^2 (\Delta - \delta/3) D}. \quad (1)$$

Provided that the self-diffusion coefficient D of the spin-bearing molecules is independent of time, it can be computed by the one-dimensional version of the Einstein–Smoluchowski equation:

$$\langle x^2 \rangle = 2Dt. \quad (2)$$

Another prerequisite for the validity of this equation is the existence of a Gaussian propagator, which is realized either for free (unbound) diffusion of simple liquids or for confined motion where the mean-squared displacements exceed the largest structural size in the system. For bulk ILs, it has been shown repeatedly that the Gaussian assumption is correct for the timescales typically accessible by PFG NMR experiments.

A deviation from these conditions can often be identified by a deviation of the signal decay from a single exponential function. Easier to observe, however, frequently is the dependence of D on the encoding time Δ . Although

this is, strictly speaking, an indication for the violation of the Gaussian assumption, it has become accepted practice to express the thus obtained quantity as a time-dependent self-diffusion coefficient, $D(\Delta)$.

Relaxation of a nuclear spin system relies on the spectrum of local magnetic field fluctuations at the position of the nucleus, which is predominantly modulated by the motions of the spin-bearing molecule itself and of its environment. In principle, one can separate this motion into a rotational contribution and a translational contribution, and the total relaxation rate is a sum of these two processes:

$$\frac{1}{T_1(\omega)} = \frac{1}{T_{1,\text{intra}}(\omega)} + \frac{1}{T_{1,\text{inter}}(\omega)}, \quad (3)$$

where $\omega = 2\pi\nu = \gamma B_0$ denotes the ^1H Larmor frequency, γ is the gyromagnetic factor, and B_0 is the magnetic flux density. $T_{1,\text{intra}}(\omega)$ expresses the intramolecular magnetic dipolar interactions among protons within the same molecule modulated by local molecular reorientations. $T_{1,\text{inter}}(\omega)$ describes the intermolecular magnetic dipolar proton interactions modulated by the translational self-diffusion of molecules.

The homonuclear spin–lattice relaxation rate for intramolecular dipolar interaction is given by the Bloembergen, Purcell, and Pound equation

$$\frac{1}{T_{1,\text{intra}}} = \frac{3}{10} \left(\frac{\mu_0}{4\pi} \right)^2 \frac{\gamma^4 \hbar^2}{r^6} [J_{\text{intra}}(\omega) + 4J_{\text{intra}}(2\omega)], \quad (4)$$

where μ_0 is the vacuum permeability, $\hbar$ is the Planck constant, and r is the internuclear distance. The spectral densities $J(\omega)$ are defined as the Fourier transform of the autocorrelation function that characterizes the dynamics of the molecules. $J_{\text{intra}}(\omega)$ corresponds to the spectral density of the rotational motion. Assuming a single rotational correlation time τ_R , $J_{\text{intra}}(\omega)$ is given as

$$J_{\text{intra}}(\omega) = \frac{\tau_R}{1 + (\omega\tau_R)^2}.$$

The homonuclear intermolecular (translational diffusion) contribution to the spin–lattice relaxation is given as^[25]

$$\frac{1}{T_{1,\text{inter}}(\omega)} = \frac{3}{10} N_H \left(\frac{\mu_0}{4\pi} \right)^2 \gamma^4 \hbar^2 [J_{\text{inter}}(\omega) + 4J_{\text{inter}}(2\omega)], \quad (5)$$

where $J_{\text{inter}}(\omega)$ corresponds to the spectral density of the translational motion and N_H is the number of nuclei per unit volume. Taking into account the simple force-free hard-sphere model for translational diffusion,^[26,27] $J_{\text{inter}}(\omega)$ is given as

$$J_{\text{inter}}(\omega) = \frac{72}{d^3} \int_0^\infty \frac{u^2}{(81 + 9u^2 - 2u^4 + u^6)[u^4 + (\omega\tau_{\text{trans}})^2]} du, \quad (6)$$

where d is the minimum distance of approach; τ_{trans} is the translational correlation time defined as $\tau_{\text{trans}} = d^2 / D_r$, where D_r is the relative translational diffusion coefficient; and u is a dimensionless integration variable.

In the ILs presented in this work, the cation contains ^1H and the anion contains ^{19}F as the most sensitive nucleus. Both nuclei possess similar gyromagnetic ratios, which gives independent information about the motion of both ions.

In an earlier study of the same ILs in bulk,^[10] self-diffusion coefficients and translational correlation times had been obtained, and it has been shown that for ^1H relaxation, the heteronuclear contribution of the anions' ^{19}F can be neglected but not in the opposite case. The main reason for this is the lower spin density of ^{19}F compared with ^1H (six vs. 15 nuclei per ion pair). In Ordikhani-Seyedlar et al.,^[10,11] the relaxation dispersion of ILs with the same cation was successfully described by a combination of rotation and translation with different activation energies well into the supercooled region.

The NMR relaxation in a porous medium follows different rules mainly by the interaction of the fluid with the walls in the porous matrix. These might act as strong relaxation sinks, in particular if they contain a significant amount of paramagnetic impurities.^[28] In systems where this is not the case, including Vycor, which was used in this study, it has been shown that the ^1H spins of nonpolar liquids show only moderate dispersion $T_1(\omega)$, whereas polar liquids such as water and alcohols, but also the aprotic acetone, follow exceptionally strong dispersion in between several kHz and MHz of Larmor frequency. The fact that ^2H in D_2O , relaxing predominantly via quadrupolar relaxation, which has the same functional dependence on spectral density as dipolar relaxation, was observed to follow the same dispersion^[5] is considered as proof of two facts: (a) Relaxation is dominated by a single-spin effect, that is, translational contributions are negligible, and (b) the local fluctuations of the magnetic fields are solely brought about by rotational motions of the molecules. This is expressed in the term RMTDs, which require a preferential orientation that the molecule assumes in contact to the surface and which is affected by the curvature of the same. Clearly, polar molecules on the polar surface of untreated glass were shown to assume preferential orientations, whereas organic surfaces may behave differently.

The process of RMTD is active irrespective of the way that molecules diffuse along it. For instance, in the non-frozen layer between a solid wall and an otherwise frozen medium, molecules are required to “slide” along the surface and follow its curvature, or roughness spectrum, directly by obeying Gaussian diffusion statistics.^[29] Their T_1 dispersion would then reflect the surface roughness via

$$T_1 \propto D^{(1-\chi)/2} \omega^{(1+\chi)/2}, \chi \geq 0, \quad (7)$$

where D is the bulk diffusion coefficient and $\chi = d_E - d_k$ is the difference between the Euclidean dimension d_E and the surface fractal dimension d_k .^[30]

If motion is restricted only on one side, on the other hand, molecules are allowed to desorb and will then diffuse in three dimensions through the bulk fluid. Nevertheless, it has been shown that the brief encounters on the surface are still the dominating effect that determines NMR relaxation behavior; in other words, isotropization in the bulk does not lead to a fast decay of the orientational autocorrelation function of the spins; rather, the memory of the repeated adsorptions in a defined orientation prolongs this autocorrelation function up to the point when the molecule has, on average, moved along the curvature of the available pore space and assumed all orientations with equal probability. Bychuk and O'Shaughnessy^[7] have predicted that the statistics of motions *on the surface* then must be non-Gaussian, in fact following the Lévy walks. In this case, the T_1 dispersion is

$$T_1 \propto D^{(1-\chi)} \omega^\chi, \chi \geq 0. \quad (8)$$

This prediction has been proven experimentally by NMR relaxometry,^[31] and the difference between Gaussian and non-Gaussian propagators could be demonstrated unequivocally in a system of completely filled pores with the bulk phase either frozen or liquid.^[30,31] The latter situation has been named “bulk-mediated surface diffusion.” It is important to realize that bulk excursions are required for establishing the non-Gaussian surface diffusion propagator and that fast—on the timescale of the Larmor frequency—exchange between surface and bulk is assumed in this case but that RMTD itself does not require this exchange. If mobilities close to the surface and in the bulk-like volume of the pore are noticeably different, slow exchange (on the timescale of T_1) would be observed by a nonexponential relaxation decay. In fact, in Chathoth et al.,^[32] two such components in neutron scattering experiments were resolved, but they report residence times as short as 10^{-11} to 10^{-10} s, inaccessible to NMR.

4 | RESULTS

The self-diffusion coefficients of cations of bulk and confined Bmim Tf2N are shown in Figure 1 in a broad temperature range, from 308 to 248 K. The corresponding echo attenuations from the ^1H signal obtained from the confined IL are shown as well, indicating single exponential decays. The encoding time Δ and the gradient pulse duration δ were set to 100 ms and 100 μs , respectively. The self-diffusion coefficients show an Arrhenius dependence with the temperature for both the bulk and confined liquids.

The results of the spin–lattice relaxation dispersion of the confined Bmim Tf2N are shown in Figure 2a,b. For comparison, the bulk T_1 values in a similar range of temperature are shown in Figure 3a,b. By comparing the data, at higher temperatures, the difference in the frequency dependence of T_1 is dramatic, and the effects of confinement are evident. However, at low temperatures, the T_1 dispersions have similar frequency dependence. The bulk T_1 dispersion data can be described^[10,17] by using Equations 5 and 6.

In order to demonstrate how the T_1 relaxation dispersion of bulk and confined ILs converges at low temperatures, cation and anion results are compared for Bmim Tf2N at three temperatures $T \leq 1.2T_g$ in Figure 4. In the case of the IL under confinement at the temperature around 198 K, it was not possible to measure with accuracy the T_1 relaxation times at frequencies below 0.1 MHz. Several factors related with sensitivity affect these measurements. Apart from the fact that the amount of sample is less than in bulk, the transversal (T_2) NMR relaxation time shortens significantly the signal decay. These two factors are crucial for a good determination of T_1 values in the millisecond range and below.

The T_1 dispersions of cation and anion molecules under confinement (Figure 2) show well-defined power-law frequency dependences. At higher temperatures, a single power law with an exponent 0.66 is observed in the complete frequency range. However, when the temperature is decreased, another power-law dependence starts to appear with an exponent close to 1.0. The two limits are shown in Figure 2. The onset of transition between these two regimes is gradually decreasing to low frequencies follow-

FIGURE 1 (Left) Cation self-diffusion coefficients of bulk (black) and confined (red) Bmim Tf2N. (Right) Echo attenuation signal of Bmim Tf2N under confinement measured using the SE-PFG–NMR experiments. NMR: nuclear magnetic resonance; SE-PFG: stimulated-echo pulsed field gradient

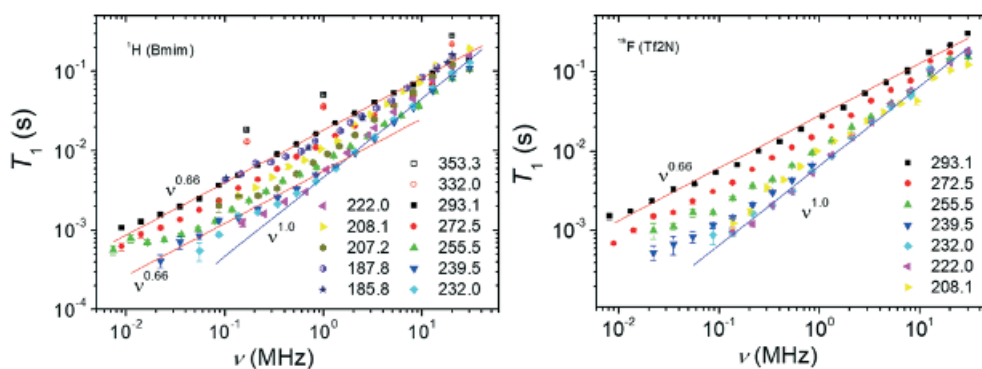
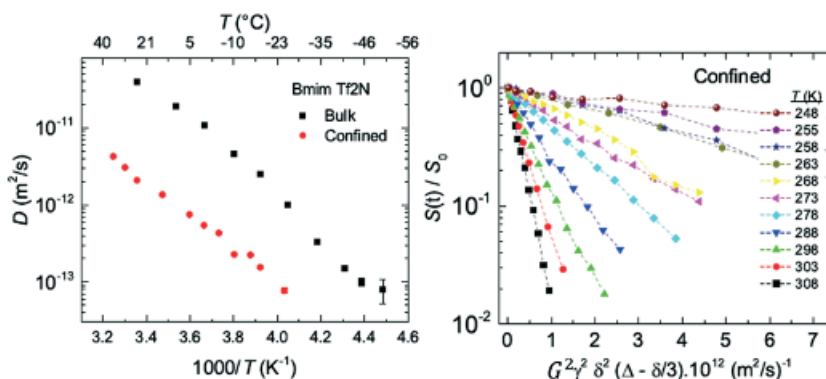


FIGURE 2 (Left) ^1H - T_1 and (right) ^{19}F - T_1 dispersions of Bmim Tf2N confined in Vycor corresponding to the cation and anion. The fittings with the solid lines are explained in the text

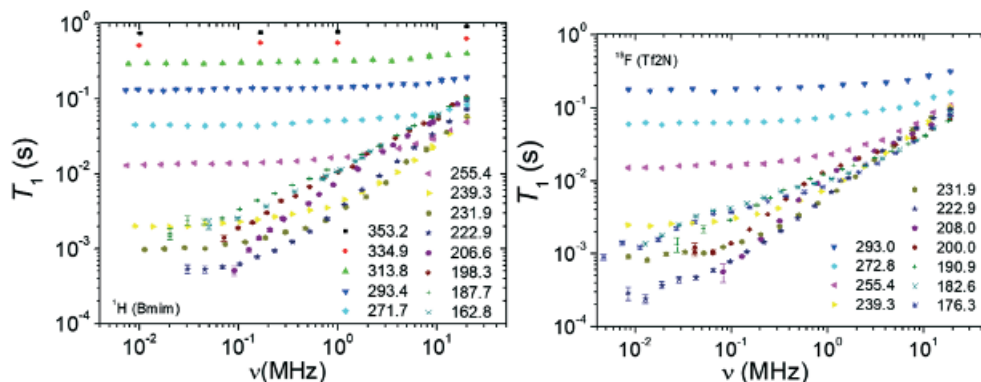


FIGURE 3 (Left) ^1H - T_1 and (right) ^{19}F - T_1 dispersions of bulk Bmim Tf2N corresponding to the cation and anion

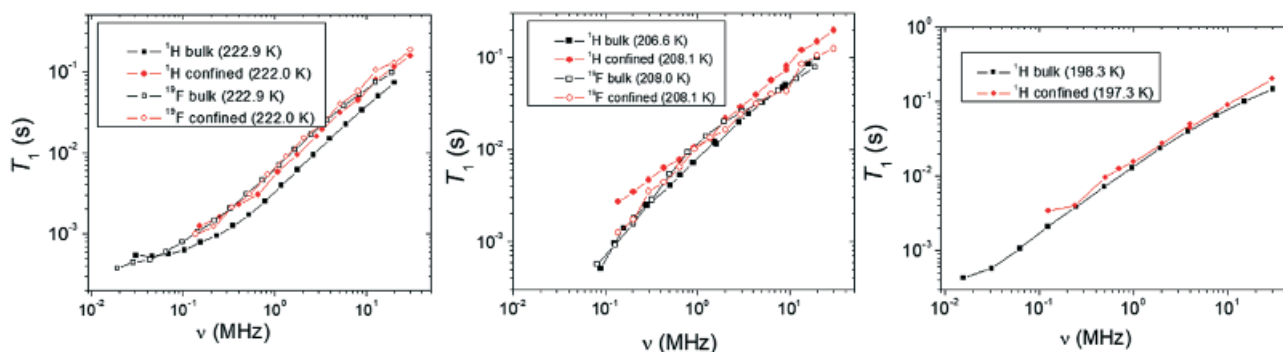


FIGURE 4 Superimposed cation and anion T_1 dispersion curves in bulk and confined Bmim Tf2N for the temperatures about 223, 208, and (only cation) 198 K

ing the decrease in temperature, but the value of the exponents of the power law remains unchanged.

The ^1H - T_1 dispersions of the IL Emim Tf2N are shown in Figure 5. The behavior of the relaxation data is qualitatively the same as that for the Bmim Tf2N IL; however, low-temperature data for bulk are not available because the sample freezes only a few degrees below its nominal melting point; that is, significant supercooling does not occur in contrast to Bmim Tf2N. The power-law fittings

are shown as well, with the same exponent as in the case of Bmim Tf2N.

For comparison, the ^1H - T_2 values of the IL Emim Tf2N both in bulk and confined in Vycor porous glass are shown in Figure 6. The ^1H - T_2 relaxation times in the bulk IL exhibit a deviation from an Arrhenius dependence at low temperature typical of glass-forming liquids, whereas for the confined IL, a linear dependence is observed.

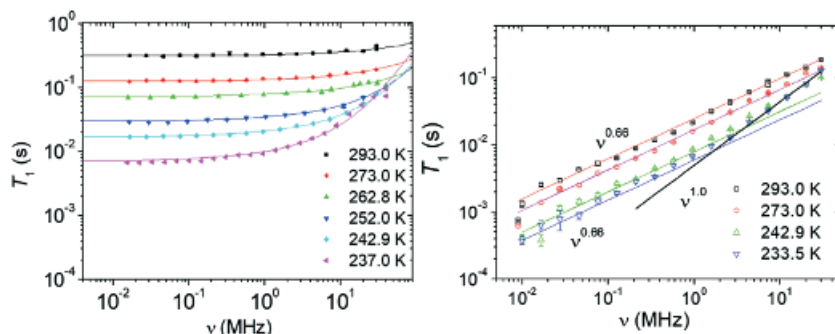


FIGURE 5 (Left) Bulk ^1H - T_1 and (right) confined ^1H - T_1 dispersions of Emim Tf2N corresponding to the cation

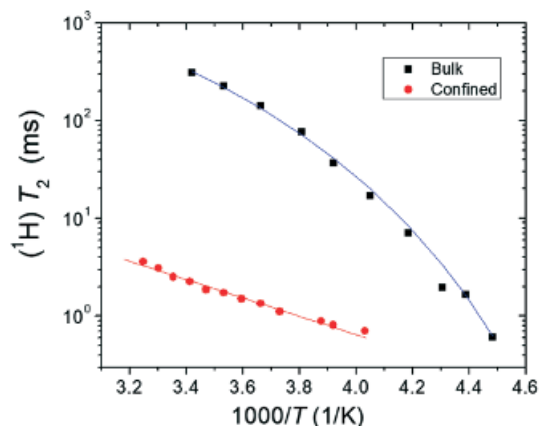


FIGURE 6 Bulk and confined T_2 relaxation values of Bmim Tf2N corresponding to the cation. The confined values are fitted with an Arrhenius temperature dependence, whereas the bulk values are fitted with a Vogel–Fulcher–Tammann model (see text)

5 | DISCUSSION

This ratio of diffusion values between bulk and confined Bmim Tf2N IL in Figure 1 shows a reduction of the bulk diffusivity of about 1:17 for all temperatures, indicating a tortuosity of around 17. This result differs to what is found in early work of different polar and nonpolar liquids in Vycor where the reduction of the bulk diffusivity was consistently found to be about 1:6 irrespective of molecular size and polarity.^[33] However, more complex liquids such as polymer melts under confinement show a much slower diffusion, but one has to consider that the characteristic size of these macromolecules, represented by the radius of gyration (R_g), is larger than the pore size and might have many interaction locations with the surface. In the case of IL, the interionic interaction and interaction with the OH-groups in the surface of the porous material may explain the slowing down in the mobility in the same way at different temperatures, by a factor of about 3, compared with tortuosity. On the other hand, from the temperature dependence in Figure 1, in the overlapping temperature range, the activation energy is actually identical on average (39 ± 2 kJ/mol).

As is mentioned above, ILs exhibit dispersion in T_1 relaxation values at different Larmor frequencies. In earlier works, the principle of RMTD has been demonstrated for simple liquids that did not possess a bulk dispersion, that is, $T_1 \approx \text{const.}$, over the entire accessible frequency range. An approach to analyze the relaxation data would be to add relaxation rates of different contributions, that is, “bulk-like” and “surface-like,” a strategy that works

well for large pores as is best described by the Brownstein–Tarr model,^[34] which is the basis of many pore-size-determination methods with NMR. However, for very small pores, the assumption of bulk-like behavior cannot be maintained anymore, because each molecule hits the surface with a time of typically $r^2/6D$, where r is the pore radius and D is the bulk self-diffusion coefficient. Assuming $r = 2$ nm and making the simplification of spherical pores in Vycor, this corresponds to 0.3 ns for water and between 10 ns and 8 μ s for the IL Bmim Tf2N in the studied temperature range. In the case of liquids that possess a pronounced T_1 dispersion already in the bulk, such as for IL at low temperatures, the situation is less obvious. In Neves et al.,^[19] Martins et al.,^[20] and Kaszynska et al.,^[21] the total relaxation rate was considered as a sum of relaxation rates based on a bulk-like component and an RMTD component; in Kruk et al.,^[22] the same was done for two standard rotational and translational components for very small (2.7-nm) pores. The choice between the two general approaches will depend on knowledge of the interaction and the orientation of the studied IL molecules close to the surface. By observing the T_1 dispersion data in Figures 2 and 3, a decomposition of the measured relaxation rate into two additive components becomes mathematically impossible at low temperatures in this study because the relaxation times T_1 in confinement are consistently longer than in the bulk, a rather unexpected result that suggests a more complicated mechanism for determining the molecular mobility from T_1 dispersion data. In this work, the relaxation data show that the dynamics responsible for the relaxation process in the ILs, from the T_1 dispersions in Figures 2 and 5 (right), can be attributed to reorientations on the surface walls. The power-law dependence of T_1 on Larmor frequency can be fitted with the RMTD model with either Equation 7 or 8. These two equations represent a different approach for describing the relaxation process as explained above. The absence of a low-frequency plateau prohibits the determination of a cutoff limit exchange, or the longest correlation time of the reorientation relaxation process (known as α process^[30]), before the molecules exchange with the bulk-like part of the liquid, far from the walls. Information about a low-frequency plateau can be estimated from the transverse relaxation time T_2 scaled by a factor of ~ 0.3 to obtain the limiting value for low field.^[6,35] From Figure 6, these values range between $T_{2,\text{low-freq}} \approx 1.2$ and 0.15 ms for the higher and lower temperatures, respectively. This limiting approximation is consistent with the low-field T_1 relaxation data from Figure 2 (right) of Bmim Tf2N confined in Vycor.

An important fact is that in the ILs studied in this work (the ^{19}F - T_1 dispersion of Emim Tf2N is not shown),

both ions have similar relaxation dispersions (Figure 2a, b) with the same power-law dependence, indicating that the ion-surface interaction is the same; in other words, these experiments show no evidence that positive and negative ions will interact differently with the surface.

Figure 3 shows a strong dependence of the bulk IL dynamics with the temperature. In Ordikhani-Seyedlar et al.,^[10] it was shown that the characteristic rotational and translational correlation times differ by several orders of magnitude in the temperature range shown in the figure. However, this is not the case when the same liquid (and Emim Tf2N) is confined (see Figures 2, 4, and 5). The exponent of the power-law frequency dependence ($\omega^{0.66}$) is not changing with temperature at low frequencies. However, when the temperature decreases, a dependence in the form of $\omega^{1.0}$ appears at high frequencies, and the transition frequency between the two slopes is shifting with temperature. In order to account for the temperature dependence, one could write Equations 7 and 8 as $T_1 = AD^\beta \omega^\mu$, where A indicates the type of interaction, independent of temperature. β and μ can be inferred from the respective Equations 7 and 8. The temperature dependence is in this case introduced by the diffusion coefficient. Taking μ as the fitting exponent value, it was found that

$$T_1 = AD^{1-\mu} \omega^\mu. \quad (9)$$

In this way, by fitting the experimental data with Equation (9), the ratio between diffusion coefficients at different temperatures is found. With the use of a reference experimental value, the diffusion data can be determined. Figure 7 shows the diffusion values determined via the scaling from Equation (9) with $\mu = 0.66$, and in comparison with the experimental values. The agreement is reasonably good, and deviations start to be seen at $T \approx 1.2T_g$. The change in slope with the exponent $\mu = 1.0$ is related with an additional relaxation process, not related with RMTD.

Figures 2, 3, and 4 show that the T_1 dispersions for both ions approach each other at low temperature. At $T \approx 1.2T_g$ and below, the frequency range of convergence covers several orders of magnitude. This is an indication that the bulk and confined dynamics of the IL, at these low temperatures where the bulk IL is deeply supercooled, becomes similar and could be described with similar characteristic correlation times. For this case, the RMTD model for the confined IL cannot be applied anymore.

The difference in the dynamical behavior at low temperatures can be seen as well from the spin-spin relaxation data in Figure 6. In the limit of $\omega\tau_c \ll 1$ (extreme narrowing), the relation $1/T_{2,\text{conf}} \propto \tau_{c,\text{conf}} \propto e^{(E/RT)}$ can be written. The fitted straight line in Figure 6 corresponds to this limit, giving an activation energy $E \approx (18 \pm 1)$ kJ/mol. The bulk values follow in the same limit, a VFT temperature dependence,

$$1/T_{2,\text{bulk}} \propto \tau_{c,\text{bulk}} \propto e^{\left[\frac{K T_0}{T - T_0}\right]}, \text{ where } T_0 \text{ usually denotes}$$

the theoretical low temperature limit to the liquid state. $T_0 = (173 \pm 12)$ K and $\kappa = (3.1 \pm 0.2)$. The constant κ is the fragility parameter, used to characterize how steeply viscosity increases upon cooling, and it can be regarded as a measure of the degree of cooperativity of motion.^[36] Liquids are called strong if the Arrhenius behavior is obeyed, whereas they are called fragile in the other case.^[36] Bmim Tf2N loses its fragility when it is under confinement in the mesoporous glass Vycor and becomes strong. In Ordikhani-Seyedlar et al.,^[10] slow rotational and translation dynamics of the bulk IL could be distinguished by analyzing the T_1 dispersion data of Figure 3. In this case, the activation energy of the rotational process of the bulk IL could be calculated for temperatures above $T \sim 1.24T_g$, with a value $E_R = (29 \pm 2)$ kJ/mol.^[10] In this work, T_2 relaxation times in the bulk IL are measured in the high-frequency limit instead (Figure 6). The rotational and translational dynamics cannot be distinguished in this case. The fitted values with a VFT

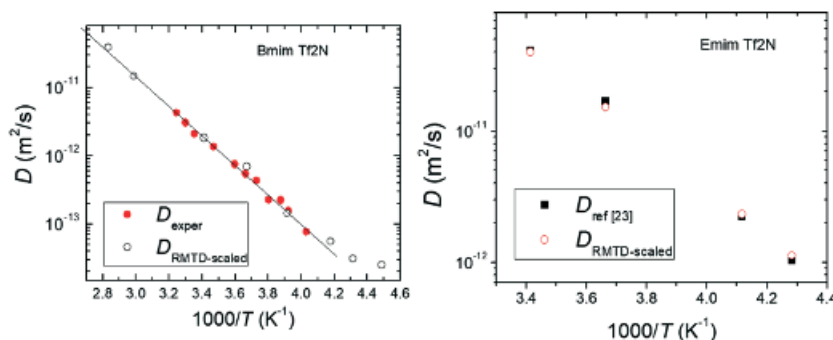


FIGURE 7 (Left) Bmim Tf2N and (right) Emim Tf2N cation diffusion coefficients calculated from relaxation data using Equation (9) (open points). The solid points correspond to the Bmim cation values from Figure 1 and to the Emim cation values obtained from Tokuda et al.^[23]

equation are reasonable, in comparison with values in Ordikhani-Seyedlar et al.^[10]; however, $\tau_{c,bulk}$ has no clear meaning in this case and can be regarded as a representative value describing the complex dynamics of the cation in the supercooled regime. The low activation energy of the confined cation molecules is an indication that the rotational dynamics is somehow facilitated inside Vycor due to the less cooperative character in its dynamics.^[37] The geometrical restrictions affecting the dynamics of the confined IL are effective when the correlation length of the structure fluctuations in the bulk liquid goes beyond the characteristic pore sizes. The correlation length decreases with the temperature; and when it becomes into the order of the porous size, the dynamics in both states of the liquid becomes similar.^[38] This is observed as well in the T_1 dispersions in Figure 4, where below $T \leq 1.2T_g$, the curves start to overlap.

6 | CONCLUSIONS

Diffusion inside porous Vycor is reduced from the bulk diffusivity by a factor that is larger than expected from the known tortuosity in this material. The factors responsible for this are interactions between ions in ILs and between the ions and the surface.

The RMTD relaxation process was shown for IL in confined environment of 4-nm pore size. This is indicated by the power-law dependence of T_1 relaxation times with the Larmor frequency with the exponent 0.66. This is in very good agreement with what is found in literature^[29] for other liquids confined in Vycor and similar porous media. A single power law is observed at high temperatures; however, a second power law is necessary to describe the relaxation results at high frequency when temperature is decreased. This can be attributed to the appearance of a different relaxation mechanism due to the onset to a more cooperative dynamics while temperature approaches $T \approx 1.2T_g$. Below this temperature, practically no difference is observed between bulk and confined dynamics in Bmim Tf₂N IL. In order to explain this observation, more research is required where several questions must be answered, such as what is the relaxation mechanism responsible for the high-frequency T_1 dispersion and why do different ions interact in a similar way with the surface of the porous material. More fundamental is how general are the results in this work about the dynamics of other bulk and confined ILs when they are deeply supercooled.

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